

University of Utah

School of Computing

Computer Vision

CS 5320/6320

Spring Semester 2016

WEB 2250 MW 1:25-2:45

Instructor: Thomas C. Henderson

Overview of Course

- [Course Objectives](#)
- [Prerequisites](#)
- [Course Description](#)
- [Software Used to Support Class](#)
- [Required Materials](#)
- [Projects and Assignments](#)
- [Class Schedule and Assignments](#)
- [Instructor](#)
- [Grading Information](#)
- [Assignment Due Time](#)
- [Assignment Late Policy](#)
- [Individual Work](#)
- [Registration](#)
- [American Disabilities Act](#)
- [Link to Course Info and Documentation](#)
- [Link to Programming Tips](#)
- [Link to Code](#)
- [Link to Lectures](#)
- [Lab Report Format](#)
- [Link to Testing and Data](#)
- [Link to Solutions](#)
- [College Guidelines](#)

Course Objectives

Computer vision (from Forsyth & Ponce):

- is an enterprise that uses statistical methods to disentangle data using models constructed with the aid of geometry, physics and learning
- obtains simple inferences from individual pixel values
- combines information from multiple images into a coherent whole
- imposes order on groups of pixels to segment them or infer shape information
- recognizes objects using geometric information or probabilistic techniques.
- has a wide variety of applications:
 - mobile robot navigation
 - image retrieval from digital libraries
 - medical image analysis
 - realistic rendering of synthetic scenes in computer graphics.

The following lists the goals for the computer vision course:

- Experience with computer vision tools (Matlab)
- Explore computer vision in terms of the issues described above
- Learn appropriate reporting techniques

Prerequisites

The prerequisite is successful completion of CS 3505, Math 2210, and Math 2270 or equivalent.

Course Description

We will work on the problems and solutions to image analysis using modern computer vision techniques.

Software Used to Support Class

Students will develop codes in Matlab.

Required Materials

We will use:

Forsyth and Ponce. Computer Vision: A Modern Approach (2nd ed), Prentice Hall, 2012 (required) and selected readings.

Assignments

There are 2 major types of assignments (use the [Lab Report Format](#) for Problem reports):

- Problem Assignments: These must be completed by the due date and turned in as PDF report files.
- Quizzes: Weekly (short) quizzes will be held during the semester.

Class Syllabus

The lectures will cover the text on the following schedule (may vary some during semester to accommodate progress):

Date	Topic	Material (Forsyth and Ponce)	Problem Assignment
January 11-27	Geometric Camera Models, Light and Shading	Chapters 1-2	Assigned: A1 , A2 , A3
February 1-17	Linear Filters, Local Image Features	Chapters 4-5	Assigned: A4 , A5 , A6
February 22 - 24	Texture	Chapter 6	Assigned: A7
February 29, March 7-9	Segmentation by Clustering	Chapter 9	Assigned: A8
March 14-16	Spring Break		
March 21 - 30	Grouping and Model Fitting	Chapter 10	Assigned: A9
April 4-13	Tracking	Chapter 11	Assigned: A10
April 18-25	Range Data Analysis	Chapter 14	Assigned: A11

Class Schedule and Assignments

The lectures and assignments will cover the text as we progress through the semester. Assignments will usually be handed out on Monday and a Wednesday after the material is covered.

Instructor

Instructor:

[Thomas C. Henderson](#), Professor

E-Mail:

tch@cs.utah.edu

Phone:

801-581-3601

Fax:

801-585-3743

Office Hours (2781 WEB): By appointment.

[N/A](#)

E-Mail:

N/A

Phone:

N/A

Fax:

N/A

Office Hours: N/A

Grading Information

The grading distribution will be as follows:

- Problems: 70%
- In-Class Tests and Participation: 30%

You are expected to make a good effort on all assignments and in-class discussion based on a careful reading of the assigned material. I will assign grade based on how reasonable your solution is given the difficulty of the assignment, the time required, and the style and content of the solution. My goal is to look at all your work, and to assign a grade based on your participation, effort and results. It's better to ask questions before and during an assignment, than to try and understand what went wrong after it's due. The proportions given above delineate how I intend to apportion the weight of the various work in the course.

Assignment Due Time

Unless otherwise stated in an assignment, all assignments will be due by classtime on the assignment due date. You should handin PDF's and code. The time that we use for an assignment is the submit time. I may ask for supporting material as well (Matlab codes, images, math analysis, etc).

Appeals Procedure

See the Code of Student Rights and Responsibilities, or the Class Schedule for more details.

Appeals of Grades and other Academic Actions

If a student believes that an academic action is arbitrary or capricious he/she should discuss the action with the involved faculty member and attempt to resolve. If unable to resolve, the student may appeal the action in accordance with the following procedure:

1. Appeal to Department Chair who should be notified in writing within 40 working days; chair must notify student of a decision with 15 days. If the faculty member or student disagrees with decision, then,
2. Appeal to Academic Appeals Committee (see flyers posted in MEB and EMCB for members of committee). See II Section D, Code of Student Rights and Responsibilities for details on Academic Appeals Committee hearings.

Assignment Late Policy

No late work is accepted.

Individual Work

The purpose of the assignments is to improve your skills at solving problems and demonstrating that you understand the class material. Collaboration with other class members is acceptable in understanding problems or software tools. For any individual assignments or work turned in, you must submit your own work. Using someone else's work or giving someone else your work is considered plagiarism and will be dealt with using standard CS and University procedures (i.e., failure of assignment and class). The SoC policy states: "As defined in the University Code of Student Rights and Responsibilities, academic misconduct includes, but is not limited to, cheating, misrepresenting one's work, inappropriately collaborating, plagiarism, and fabrication or falsification of information. It also includes facilitating academic misconduct by intentionally helping or attempting to help a student to commit an act of academic misconduct. A primary example of academic misconduct would be submitting as one's own, work that is from an outside source." (See [cheating_policy.pdf](#) and [SoC_ack_form.pdf](#) in Link to Class Info and Docs.)

Registration

See university web page for the full academic calendar ([Calendar web page](#)). See the university web page for a copy of the withdraw guidelines, well, or see the [Student Code](#).

See the college web page for more [Guidelines](#).

The University conforms to all standards of the ADA. If you wish to qualify for exemptions under this act, notify the Center for Disabled Student Services, 160 Union.

The University of Utah seeks to provide equal access to its programs, services and activities for people with disabilities. If you will need accommodations in the class, reasonable prior notice needs to be given to the Center for Disability Services, 162 Olpin Union Building, 581-5000 (V/TDD). CDS will work with you and the instructor to make arrangements for accommodations.

All written information in this course can be made available in alternative format with prior notification to the Center for Disability Services.
